F6 Capstone Project – Progress Update

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1. Project title

Explainable Machine Learning for Screening Kidney Disease Markers Among Adults with Diabetes Using NHANES Clinical Data

Healthcare problem: Kidney abnormalities among adults with diabetes may remain undetected. Identifying individuals with abnormal kidney markers using routinely collected clinical and demographic variables could support preliminary screening and prioritization for further clinical assessment.

Intended users: Healthcare professionals, clinical researchers and public health screening teams.

Decision supported: The proposed system aims to identify individuals who may require further kidney-function assessment based on their clinical and demographic characteristics. It is intended as a screening-support tool, not a diagnostic system.

2. Objectives

The primary objective is to develop an explainable machine learning framework to identify kidney abnormality markers among adults with diabetes using NHANES clinical and demographic data.

Specific objectives:

1.Merge NHANES demographic, diabetes, biochemical, anthropometric and blood pressure data.

2.Develop a binary kidney-marker outcome based on estimated glomerular filtration rate (eGFR) and urine albumin-to-creatinine ratio (UACR).

3.Perform exploratory data analysis, preprocessing, and feature selection to identify the relevant predictors.

4.Develop baseline and advanced machine learning algorithms for predicting kidney marker status.

5.Use explainability techniques, focusing on SHAP, to understand important predictors and model decisions.

6.Create an interactive dashboard for communicating predictions and feature contributions.

Analytical target and unit of analysis:

- **Unit of analysis:** Individual NHANES participant aged 20 years or older meeting the study's operational diabetes definition and examination eligibility criteria.
- **Study population:** Participants reporting a diabetes diagnosis (DIQ010 = 1) OR having HbA1c ≥ 6.5%.
- **Primary outcome:** Binary kidney-marker status.
    - Positive (1): eGFR < 60 mL/min/1.73 m² OR UACR ≥ 30 mg/g.
    - Negative (0): Both measurements are available and within the defined normal ranges.
- **eGFR calculation:** Race-free 2021 CKD-EPI creatinine equation.
- **Time window:** NHANES 2017–March 2020 pre-pandemic survey cycle. This is a cross-sectional analysis, not a longitudinal prediction study.

Proposed baseline and evaluation:

- **Baseline model:** Logistic Regression.
- **Additional models:** Random Forest and XGBoost.
- **Evaluation metrics:** Accuracy, precision, recall (sensitivity), specificity, F1-score, ROC-AUC and PR-AUC.
- **Explainability:** SHAP feature importance and individual-level explanations.
- **Validation strategy:** Stratified 80:20 training/testing split with a fixed random seed.

Proposed dashboard/output:

An interactive Streamlit dashboard displaying kidney-marker risk predictions, relevant clinical and demographic features, model performance, and SHAP-based explanations. The dashboard will support preliminary screening and interpretation rather than clinical diagnosis.

3. Work done

Dataset source and accessibility:

The project uses publicly accessible data from the National Health and Nutrition Examination Survey (NHANES), conducted by the US National Center for Health Statistics (NCHS), for the 2017–March 2020 pre-pandemic cycle.

Dataset source:
https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?cycle=2017-2020

The required datasets were downloaded and integrated using the participant identifier (SEQN).

Key variables integrated:

Data category	Variables
Demographics	Age, sex, race/ethnicity
Diabetes	DIQ010, DIQ050, DIQ070, DIQ160
Glycaemic control	HbA1c (LBXGH)
Kidney measurements	Serum creatinine (LBXSCR), urine albumin (URXUMA), urine creatinine (URXUCR), UACR (URDACT)
Anthropometric	BMI (BMXBMI), waist circumference (BMXWAIST)
Blood pressure	Systolic and diastolic BP
Lifestyle	Smoking questionnaire variables

Completed data preparation and analysis:

1. Downloaded and integrated eight NHANES datasets using SEQN. The merged dataset contained 15,560 participants.
2. Restricted the study population to adults aged 20 years and above.

3. Examined diabetes questionnaire responses and HbA1c values to establish the operational diabetes cohort definition.
4. Restricted the cohort to examined adults and applied pregnancy-related exclusions.
5. Established the final eligible cohort of 1,601 participants.
6. Implemented the 2021 CKD-EPI race-free equation to calculate eGFR.
7. Defined the kidney-marker outcome using eGFR and UACR.
8. Identified 95 participants without sufficient measurements to assign the kidney-marker outcome.
9. Created the final labelled modelling dataset containing 1,506 participants:
    - Kidney marker negative: 855.
    - Kidney marker positive: 651.
10. Performed missingness checks, questionnaire recoding, smoking-status derivation and feature engineering.
11. Conducted exploratory data analysis, including descriptive statistics, skewness analysis, outcome-wise histograms, group comparisons and correlation analysis.
12. Identified strong correlation between BMI and waist circumference (r = 0.89), informing the proposed feature-selection strategy.
13. Created a stratified training/testing split:

- Training: 1,204 participants.
- Testing: 302 participants.
- No participant-ID overlap between the two sets.

Preliminary findings:

- The kidney-marker-positive proportion in the labelled dataset is approximately 43.2%.
- The unadjusted kidney-marker rate increases across age groups, from 28% among participants aged 20–44 to 54% among those aged 70 and above in the training set.
- Higher HbA1c groups show higher observed kidney-marker rates.
- BMI and waist circumference demonstrate substantial correlation.
- Missing questionnaire responses and skip patterns were examined to avoid incorrectly classifying unavailable responses as negative responses.

Evidence of completed work:

- NHANES source: CDC NHANES 2017–March 2020.
- Data integration and cohort preparation: `01_data_audit.ipynb`, `02_data_integration.ipynb` and `03_cohort_and_outcome.ipynb`.
- Exploratory analysis: `04_eda.ipynb`.
- Generated datasets: `cohort_labeled.csv`, `cohort_features.csv`, `train.csv` and `test.csv`.
- Preliminary visual evidence: predictor distribution histograms and correlation analysis.

4 Limitations and next steps

The study is cross-sectional, and a single abnormal measurement cannot establish chronic kidney disease, which requires evidence of persistence. The operational diabetes definition includes either self-reported diabetes or a single HbA1c measurement ≥ 6.5%; this is a study-specific cohort definition rather than confirmed clinical diagnosis. The current EDA is unweighted and does not estimate nationally representative US prevalence. Missing measurements also limit the number of participants available for modelling.

5. Planned next steps:

- Complete preprocessing and model development.
- Train and compare Logistic Regression, Random Forest and XGBoost.
- Evaluate model performance on the held-out test set.
- Perform SHAP-based model interpretation.
- Develop and integrate the Streamlit dashboard.
- Complete the final report, limitations and documentation.